SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Scenario-Based Task (High)
Assessment Tool Weightage: 35%

Dr HEMAVATHI

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Given the scenario of a School Management System, identify relations and write SQL to list all students with their class teacher and grade.	BL3 – Apply	**5**
2	A hospital wants to track patient appointments. Design the relational schema and write SQL to find doctors with more than 10 appointments this month.	BL3 – Apply	**5**
3	In an e-commerce scenario, write SQL queries using sub-queries to find products that have never been ordered.	BL3 – Apply	**5**
4	For a university database, write relational algebra expressions to find students enrolled in both 'DBMS' and 'OS' courses.	BL3 – Apply	**5**
5	Given a payroll scenario, create a view showing employees whose salary is below department average and write a query to update their salary by 10%.	BL6 – Create	**5**
6	In a banking scenario, apply JOIN operations to retrieve customer account details, transaction history, and branch information.	BL3 – Apply	**5**

7	A retail store needs daily sales summary. Write SQL with GROUP BY and HAVING to report total sales per category exceeding Rs. 10,000.	BL3 – Apply	**5**
8	For a library system scenario, construct tuple relational calculus expressions to find members who borrowed books from all genres.	BL3 – Apply	**5**
9	Given an inventory management scenario, define CHECK and FOREIGN KEY constraints and demonstrate their enforcement through SQL.	BL3 – Apply	**5**
10	Design a relational schema for an HR system and create materialized views for monthly attendance and performance summaries.	BL6 – Create	**5**

Code of Conduct:

I S KUSHA LAKSHMI (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S KUSHA LAKSHMI 192571074

1	Given the scenario of a School Management System, identify relations and write SQL to list all students with their class teacher and grade.

```
mysql> SELECT * FROM Teachers;
+-----------+-------------+-------------+
| TeacherID | TeacherName | Subject     |
+-----------+-------------+-------------+
|       101 | Anitha      | Mathematics |
|       102 | Ramesh      | Science     |
|       103 | Priya       | English     |
+-----------+-------------+-------------+
3 rows in set (0.02 sec)

mysql> SELECT * FROM Classes;
+---------+-----------+-----------+
| ClassID | ClassName | TeacherID |
+---------+-----------+-----------+
|       1 | Class A   |       101 |
|       2 | Class B   |       102 |
|       3 | Class C   |       103 |
+---------+-----------+-----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Students;
+-----------+-------------+-------+---------+
| StudentID | StudentName | Grade | ClassID |
+-----------+-------------+-------+---------+
|      1001 | Rahul       | A     |       1 |
|      1002 | Sneha       | B     |       2 |
|      1003 | Arjun       | A     |       1 |
|      1004 | Meena       | A     |       3 |
|      1005 | Kavin       | B     |       2 |
+-----------+-------------+-------+---------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     S.StudentID,
    ->     S.StudentName,
    ->     C.ClassName,
    ->     T.TeacherName,
    ->     S.Grade
    -> FROM Students S
    -> JOIN Classes C
    -> ON S.ClassID = C.ClassID
    -> JOIN Teachers T
    -> ON C.TeacherID = T.TeacherID;
+-----------+-------------+-----------+-------------+-------+
| StudentID | StudentName | ClassName | TeacherName | Grade |
+-----------+-------------+-----------+-------------+-------+
|      1001 | Rahul       | Class A   | Anitha      | A     |
|      1003 | Arjun       | Class A   | Anitha      | A     |
|      1002 | Sneha       | Class B   | Ramesh      | B     |
|      1005 | Kavin       | Class B   | Ramesh      | B     |
|      1004 | Meena       | Class C   | Priya       | A     |
+-----------+-------------+-----------+-------------+-------+
5 rows in set (0.01 sec)
```

QUES 2

```
mysql> SELECT * FROM Doctors;
+----------+------------+----------------+
| DoctorID | DoctorName | Specialization |
+----------+------------+----------------+
|      101 | Dr. Kumar  | Cardiology     |
|      102 | Dr. Priya  | Neurology      |
|      103 | Dr. Ravi   | Orthopedics    |
+----------+------------+----------------+
3 rows in set (0.01 sec)

mysql> SELECT * FROM Patients;
+-----------+-------------+------+--------+
| PatientID | PatientName | Age  | Gender |
+-----------+-------------+------+--------+
|       201 | Arun        |   22 | Male   |
|       202 | Meena       |   25 | Female |
|       203 | Rahul       |   30 | Male   |
|       204 | Sneha       |   27 | Female |
|       205 | Kavin       |   35 | Male   |
+-----------+-------------+------+--------+
5 rows in set (0.01 sec)

mysql> SELECT * FROM Appointments;
+---------------+-----------+----------+-----------------+
| AppointmentID | PatientID | DoctorID | AppointmentDate |
+---------------+-----------+----------+-----------------+
|             1 |       201 |      101 | 2026-07-01      |
|             2 |       202 |      101 | 2026-07-02      |
|             3 |       203 |      101 | 2026-07-03      |
|             4 |       204 |      101 | 2026-07-04      |
|             5 |       205 |      101 | 2026-07-05      |
|             6 |       201 |      101 | 2026-07-06      |
|             7 |       202 |      101 | 2026-07-07      |
|             8 |       203 |      101 | 2026-07-08      |
|             9 |       204 |      101 | 2026-07-09      |
|            10 |       205 |      101 | 2026-07-10      |
|            11 |       201 |      101 | 2026-07-11      |
|            12 |       202 |      102 | 2026-07-12      |
|            13 |       203 |      102 | 2026-07-13      |
|            14 |       204 |      103 | 2026-07-14      |
+---------------+-----------+----------+-----------------+
14 rows in set (0.00 sec)
```

QUES 3

```
mysql> SELECT * FROM Products;
+-----------+-------------+-------------+----------+
| ProductID | ProductName | Category    | Price    |
+-----------+-------------+-------------+----------+
|       201 | Laptop      | Electronics | 65000.00 |
|       202 | Mouse       | Electronics |   800.00 |
|       203 | Keyboard    | Electronics |  1500.00 |
|       204 | Headphones  | Electronics |  2500.00 |
|       205 | Printer     | Electronics | 12000.00 |
+-----------+-------------+-------------+----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM OrderDetails;
+---------------+---------+-----------+----------+
| OrderDetailID | OrderID | ProductID | Quantity |
+---------------+---------+-----------+----------+
|             1 |     301 |       201 |        1 |
|             2 |     301 |       202 |        2 |
|             3 |     302 |       203 |        1 |
|             4 |     303 |       201 |        1 |
+---------------+---------+-----------+----------+
4 rows in set (0.00 sec)

mysql> SELECT ProductID,
    ->        ProductName,
    ->        Category,
    ->        Price
    -> FROM Products
    -> WHERE ProductID NOT IN
    -> (
    ->     SELECT ProductID
    ->     FROM OrderDetails
    -> );
+-----------+-------------+-------------+----------+
| ProductID | ProductName | Category    | Price    |
+-----------+-------------+-------------+----------+
|       204 | Headphones  | Electronics |  2500.00 |
|       205 | Printer     | Electronics | 12000.00 |
+-----------+-------------+-------------+----------+
2 rows in set (0.02 sec)
```

QUES 4

```
mysql> SELECT DISTINCT S.StudentID, S.StudentName
    -> FROM Students S
    -> JOIN Enrollments E1
    -> ON S.StudentID = E1.StudentID
    -> JOIN Courses C1
    -> ON E1.CourseID = C1.CourseID
    -> JOIN Enrollments E2
    -> ON S.StudentID = E2.StudentID
    -> JOIN Courses C2
    -> ON E2.CourseID = C2.CourseID
    -> WHERE C1.CourseName = 'DBMS'
    -> AND C2.CourseName = 'OS';
+-----------+-------------+
| StudentID | StudentName |
+-----------+-------------+
|       101 | Rahul       |
|       104 | Meena       |
+-----------+-------------+
2 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Students;
+-----------+-------------+------------+
| StudentID | StudentName | Department |
+-----------+-------------+------------+
|       101 | Rahul       | CSE        |
|       102 | Sneha       | CSE        |
|       103 | Arjun       | IT         |
|       104 | Meena       | ECE        |
|       105 | Kavin       | CSE        |
+-----------+-------------+------------+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Courses;
+----------+------------+---------+
| CourseID | CourseName | Credits |
+----------+------------+---------+
|      201 | DBMS       |       4 |
|      202 | OS         |       4 |
|      203 | Java       |       3 |
|      204 | Networks   |       3 |
+----------+------------+---------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Enrollments;
+--------------+-----------+----------+------------+
| EnrollmentID | StudentID | CourseID | Semester   |
+--------------+-----------+----------+------------+
|            1 |       101 |      201 | Semester 3 |
|            2 |       101 |      202 | Semester 3 |
|            3 |       102 |      201 | Semester 3 |
|            4 |       103 |      202 | Semester 3 |
|            5 |       104 |      201 | Semester 3 |
|            6 |       104 |      202 | Semester 3 |
|            7 |       105 |      203 | Semester 3 |
+--------------+-----------+----------+------------+
7 rows in set (0.00 sec)
```

QUES 5

```
mysql> SELECT EmployeeID,
    ->        EmployeeName,
    ->        Department,
    ->        Salary
    -> FROM Employees;
+------------+--------------+------------+----------+
| EmployeeID | EmployeeName | Department | Salary   |
+------------+--------------+------------+----------+
|        101 | Rahul        | IT         | 49500.00 |
|        102 | Sneha        | IT         | 60000.00 |
|        103 | Arjun        | IT         | 55000.00 |
|        104 | Meena        | HR         | 33000.00 |
|        105 | Kavin        | HR         | 40000.00 |
|        106 | Priya        | HR         | 50000.00 |
|        107 | Ramesh       | Finance    | 38500.00 |
|        108 | Anitha       | Finance    | 45000.00 |
+------------+--------------+------------+----------+
8 rows in set (0.00 sec)

mysql> SELECT * FROM BelowDeptAverage;
+------------+--------------+------------+----------+
| EmployeeID | EmployeeName | Department | Salary   |
+------------+--------------+------------+----------+
|        101 | Rahul        | IT         | 49500.00 |
|        104 | Meena        | HR         | 33000.00 |
|        105 | Kavin        | HR         | 40000.00 |
|        107 | Ramesh       | Finance    | 38500.00 |
+------------+--------------+------------+----------+
4 rows in set (0.00 sec)

mysql> UPDATE Employees
    -> SET Salary = Salary * 1.10
    -> WHERE EmployeeID IN
    -> (
    ->     SELECT EmployeeID
    ->     FROM
    ->     (
    ->         SELECT EmployeeID
    ->         FROM BelowDeptAverage
    ->     ) AS Temp
    -> );
Query OK, 4 rows affected (0.02 sec)
Rows matched: 4  Changed: 4  Warnings: 0

mysql> SELECT * FROM Employees;
+------------+--------------+------------+----------+
| EmployeeID | EmployeeName | Department | Salary   |
+------------+--------------+------------+----------+
|        101 | Rahul        | IT         | 54450.00 |
|        102 | Sneha        | IT         | 60000.00 |
|        103 | Arjun        | IT         | 55000.00 |
|        104 | Meena        | HR         | 36300.00 |
|        105 | Kavin        | HR         | 44000.00 |
|        106 | Priya        | HR         | 50000.00 |
|        107 | Ramesh       | Finance    | 42350.00 |
|        108 | Anitha       | Finance    | 45000.00 |
+------------+--------------+------------+----------+
8 rows in set (0.00 sec)
```

QUES 6

```
mysql> SELECT
    ->     c.CustomerID,
    ->     c.CustomerName,
    ->     a.AccountID,
    ->     a.AccountType,
    ->     a.Balance
    -> FROM Customers c
    -> INNER JOIN Accounts a
    -> ON c.CustomerID = a.CustomerID;
+------------+--------------+-----------+-------------+----------+
| CustomerID | CustomerName | AccountID | AccountType | Balance  |
+------------+--------------+-----------+-------------+----------+
|          1 | Arun         |       101 | Savings     | 50000.00 |
|          1 | Arun         |       103 | Current     | 25000.00 |
|          2 | Priya        |       102 | Current     | 75000.00 |
|          3 | Rahul        |       104 | Savings     | 60000.00 |
+------------+--------------+-----------+-------------+----------+
4 rows in set (0.00 sec)

mysql>
```

QUES 7

```
mysql> SELECT
    ->     e.EmployeeID,
    ->     e.EmployeeName,
    ->     d.DepartmentName,
    ->     e.Salary
    -> FROM Employees e
    -> INNER JOIN Departments d
    -> ON e.DepartmentID = d.DepartmentID;
+------------+--------------+----------------+----------+
| EmployeeID | EmployeeName | DepartmentName | Salary   |
+------------+--------------+----------------+----------+
|        102 | Priya        | HR             | 50000.00 |
|        101 | Ravi         | IT             | 45000.00 |
|        103 | Kiran        | IT             | 60000.00 |
|        104 | Sneha        | Finance        | 55000.00 |
+------------+--------------+----------------+----------+
4 rows in set (0.00 sec)
```

QUES 8

```
mysql> SELECT
    -> COUNT(*) AS TotalEmployees,
    -> SUM(Salary) AS TotalSalary,
    -> AVG(Salary) AS AverageSalary,
    -> MAX(Salary) AS HighestSalary,
    -> MIN(Salary) AS LowestSalary
    -> FROM Employees;
+----------------+-------------+---------------+---------------+--------------+
| TotalEmployees | TotalSalary | AverageSalary | HighestSalary | LowestSalary |
+----------------+-------------+---------------+---------------+--------------+
|              5 |   250000.00 |  50000.000000 |      60000.00 |     40000.00 |
+----------------+-------------+---------------+---------------+--------------+
1 row in set (0.01 sec)
```

QUES 9

```
mysql> SELECT
    ->     Department,
    ->     COUNT(*) AS TotalEmployees,
    ->     AVG(Salary) AS AverageSalary
    -> FROM Employees
    -> GROUP BY Department
    -> HAVING AVG(Salary) > 50000;
+------------+----------------+---------------+
| Department | TotalEmployees | AverageSalary |
+------------+----------------+---------------+
| HR         |              2 |  51000.000000 |
| Finance    |              2 |  56500.000000 |
+------------+----------------+---------------+
2 rows in set (0.01 sec)
```

QUES 10

```
mysql> SELECT * FROM HighSalaryEmployees;
+------------+--------------+------------+----------+
| EmployeeID | EmployeeName | Department | Salary   |
+------------+--------------+------------+----------+
|        103 | Kiran        | IT         | 60000.00 |
|        104 | Sneha        | Finance    | 55000.00 |
+------------+--------------+------------+----------+
2 rows in set (0.00 sec)
```